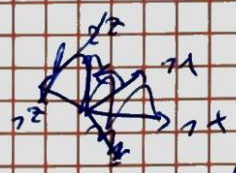
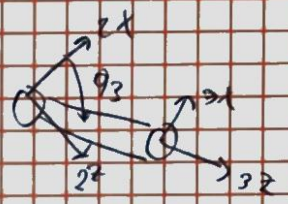
$$R_{1I} = \begin{pmatrix} -s(q_1) & c(q_1) & 0 \\ c(q_1) & s(q_1) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

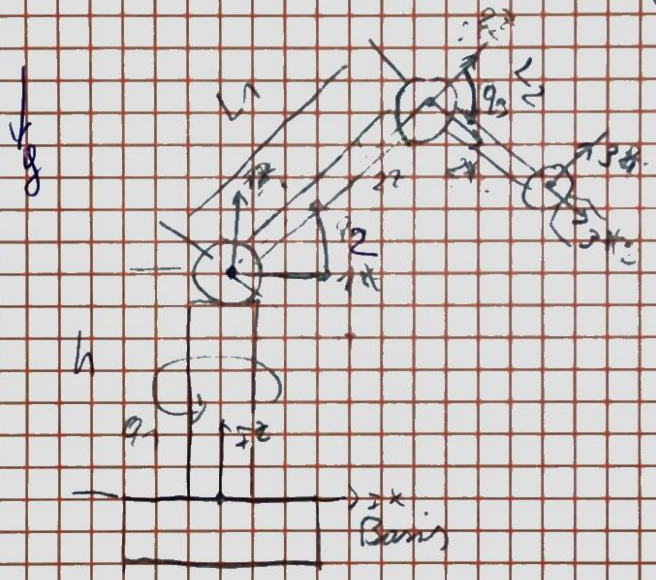
$$R_{21} = \begin{pmatrix} -s(q_2) & 0 & c(q_2) \\ 0 & 1 & 0 \\ c(q_2) & 0 & s(q_2) \end{pmatrix}$$

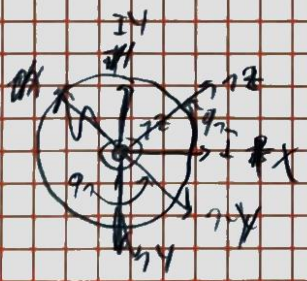


$$R_{32} = \begin{pmatrix} -s(q_3) & 0 & c(q_3) \\ 0 & 1 & 0 \\ c(q_3) & 0 & s(q_3) \end{pmatrix}$$

$$q = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} \qquad {}_I r_1 = \begin{pmatrix} 0 \\ 0 \\ h \end{pmatrix} \qquad {}_2 r_2 =$$



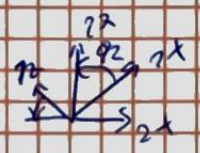




$$R_{1I} = \begin{pmatrix} c(q_1) & -s(q_1) & 0 \\ s(q_1) & c(q_1) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

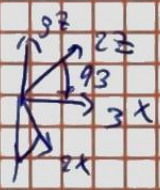
$+q_1$

$$R_{21} = \begin{pmatrix} +s(q_2) & 0 & -c(q_2) \\ 0 & 1 & 0 \\ c(q_2) & 0 & s(q_2) \end{pmatrix}$$



$-q_2$

$$R_{32} = \begin{pmatrix} -s(q_3) & 0 & c(q_3) \\ 0 & 1 & 0 \\ -c(q_3) & 0 & -s(q_3) \end{pmatrix}$$



$$R_{2I} = R_{21} R_{1I} = \begin{pmatrix} s(q_2)c(q_1) & -s(q_2)s(q_1) & -c(q_2) \\ s(q_1) & c(q_1) & 0 \\ c(q_2)c(q_1) & -c(q_2)s(q_1) & s(q_2) \end{pmatrix}$$

$$R_{3I} = R_{32} R_{21} R_{1I} = \begin{pmatrix} c(q_1)c(q_2+q_3) & -s(q_1)c(q_2+q_3) & s(q_2+q_3) \\ s(q_1) & c(q_1) & 0 \\ -c(q_1)s(q_2+q_3) & s(q_1)s(q_2+q_3) & c(q_2+q_3) \end{pmatrix}$$

$$^{I}r_{G_1} = \begin{pmatrix} 0 \\ 0 \\ h \end{pmatrix} + R_{I2} \begin{pmatrix} 0 \\ 0 \\ L_1 \end{pmatrix} + R_{I3} \begin{pmatrix} L_2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} L_1 c(q_2)c(q_1) \\ L_1 - c(q_2)s(q_1) \\ L_1 s(q_2) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ h \end{pmatrix} + \begin{pmatrix} c(q_1)c(q_2+q_3) \\ -s(q_1)c(q_2+q_3) \\ s(q_2+q_3) \end{pmatrix} L_2$$

$$^{I}r_{G} = \begin{pmatrix} L_1 c(q_2)c(q_1) + L_2 c(q_1)c(q_2+q_3) \\ -L_1 c(q_2)s(q_1) - L_2 s(q_1)c(q_2+q_3) \\ L_1 s(q_2) + h + L_2 s(q_2+q_3) \end{pmatrix}$$